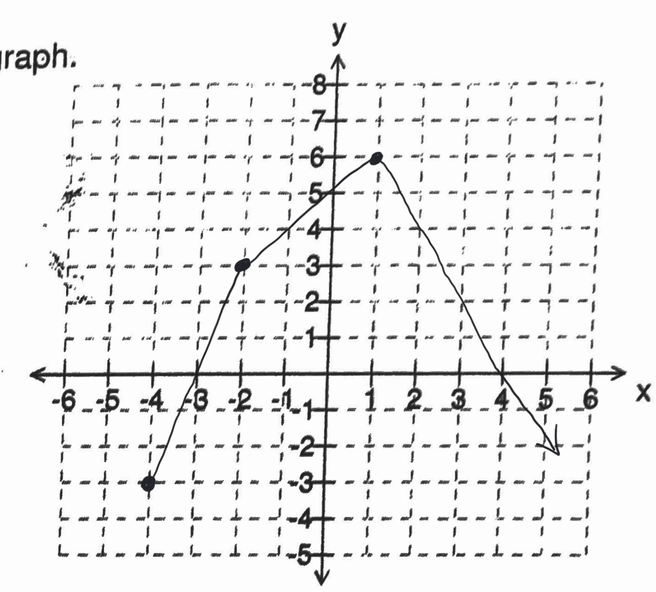
4. On the given axes draw the following piece-wise linear graph.

$$y = \begin{cases} 3x+9, & -4 \le x < -2 \\ x+5, & -2 \le x < 1 \\ 8-2x, & x \ge 1 \end{cases}$$



$y = 3x + 9$

when $x = -4$, $y = -3$

$x = -2$, $y = 3$

$y = x + 5$

when $x = -2$, $y = 3$

$x = 1$, $y = 6$

$y = 8 - 2x$ when $x = 1$, $y = 6$

$x = 2$, $y = 4$

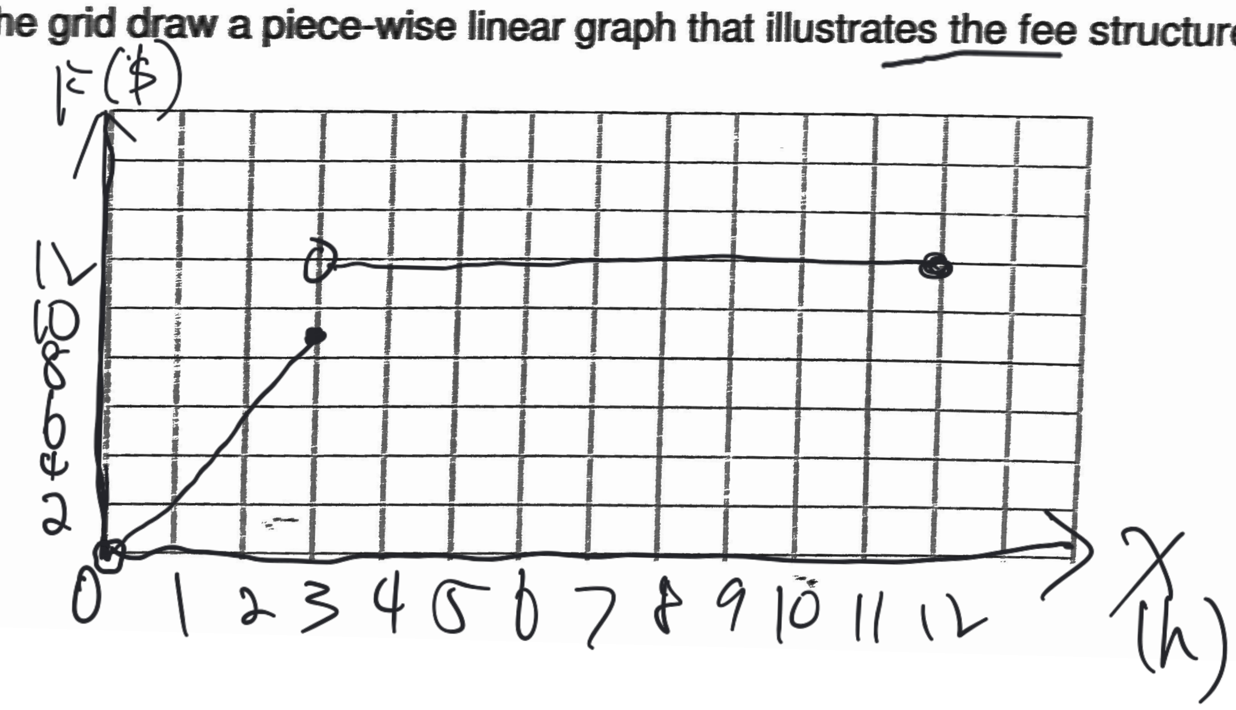
The piece-wise equation shows the weekday parking fees for a shopping-centre parking lot for up to 12 hours. The parking lot is open from 8:00 a.m. to 8:00 p.m.

$$F = \begin{cases} 3x, & 0 < x \leq 3 \\ 12, & 3 < x \leq 12 \end{cases}$$ where F is the total fee charged and x is the number of parking hours.

(a) Which description best represents the given piece-wise equation?

(i) The parking lot charges \$3 for the first 3 hours and a maximum of \$12 per day.
(ii) The parking lot charges \$3 per hour for the first 3 hours and \$12 for 3 hours after this.
(iii) The parking lot charges \$3 per hour for the first 3 hours and \$12 after this.
(iv) The parking lot charges \$3 per hour for the first 3 hours and a maximum of \$12 per day.

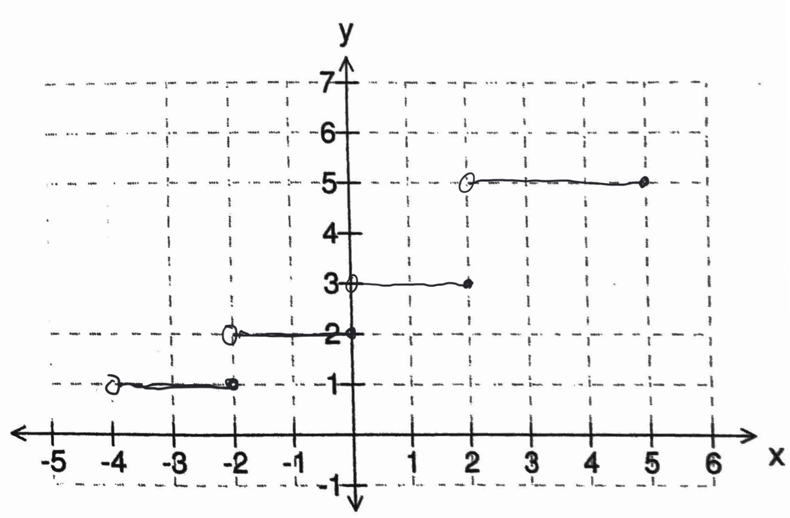
(b) On the grid draw a piece-wise linear graph that illustrates the fee structure for each weekday.



(a) On the given axes sketch the graph of

$$y = \begin{cases} 1, & -4 < x \leq -2 \\ 2, & -2 < x \leq 0 \\ 3, & 0 < x \leq 2 \\ 5, & 2 < x \leq 5 \end{cases}$$

(b) Comment on the graph you have drawn.



(a) On the given axes sketch the graph of

$$\begin{cases} 6, & -5 \leq x < -3 \end{cases}$$

y